

Chapter 2: Kinematics

Distance (definition)

- Distance (from starting point) is the shortest length between the object and the starting point.
- Scalar quantity, SI unit metre (m)

Distance Travelled (definition)

- Distance travelled is the length of path taken by an object between two points.
- ★ Distance travelled never decreases. It either remains constant or increases.

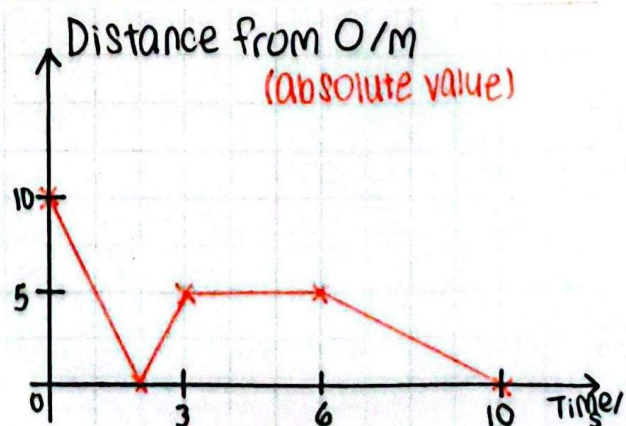
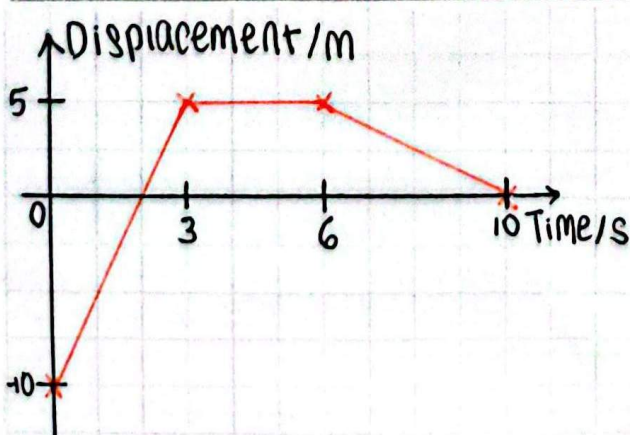
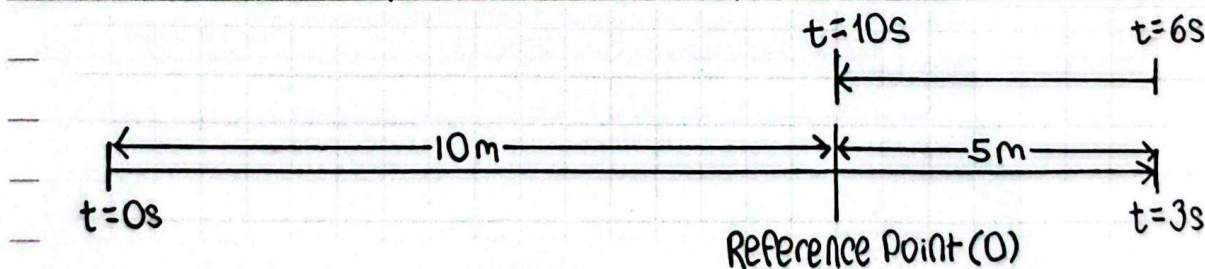
Displacement (definition)

- Displacement is the shortest distance between the object and the starting point in a specified direction.
- Vector quantity, SI unit metre (m)
- Examples:

70 km to the right of reference point

50 km to the left of reference point

Distance-Time Graph vs Displacement-Time Graph



Speed (definition)

- Speed is the distance moved per unit time.
- Average speed = Distance Travelled / Time Taken
- SI unit $\rightarrow$ m/s; scalar quantity

Converting m/s and km/h

$$1 \text{ km/h} = \frac{1000}{3600} \text{ m/s}$$

$$1 \text{ m/s} = \frac{3600}{1000} \text{ km/h}$$

Speed of sound $\rightarrow$ 330 m/s

Speed of light in vacuum $\rightarrow 3.0 \times 10^8 \text{ m/s}$

Average speed (definition)

- Average speed is the average distance travelled per unit time.
- "Between 0s to 2s, what is average speed?"

Instantaneous speed (definition)

- Instantaneous speed is the speed of an object at a particular instant. "At $t=1\text{s}$, ~~average~~ speed = ?"

Uniform speed (definition)

- Uniform speed is the constant change in the distance travelled by an object for every unit of time. (Linear graph)

Velocity (definition)

- Velocity is the rate of change in the displacement

$$\text{Velocity} = \frac{\text{Change in Displacement}}{\text{Time Taken}}$$

$$= \frac{\text{Final Displacement} - \text{Initial Displacement}}{\text{Time Taken}}$$

- Velocity is a vector quantity.
- Unit $\rightarrow$ m/s

- When describing velocity, the negative sign is used to find out the direction of velocity.

* Negative sign has no effect on the magnitude.

e.g. Object A $\rightarrow -30\text{m/s}$, Object B $\rightarrow -40\text{m/s}$, Object A and B are moving in the same direction, but Object B is faster.

e.g. Object A $\rightarrow -40\text{m/s}$, Object B $\rightarrow 20\text{m/s}$, they are moving in opposite directions.

Average velocity (definition)

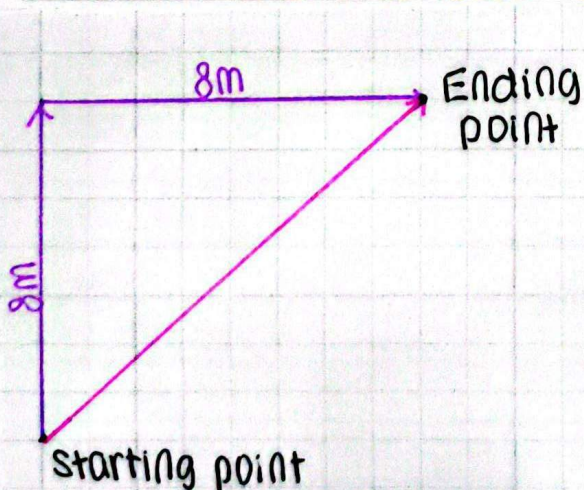
- Average velocity is the average change in displacement per unit time.

Instantaneous velocity (definition)

- Instantaneous velocity is the velocity of an object at a particular instant.

Speed vs Velocity

Example: An object moves 8m due north then 8m due east.
The time taken is 10s.



1. Average Speed

Total distance travelled $\rightarrow 8+8=16\text{m}$

Average speed $\rightarrow 16 \div 10 = 1.6\text{m/s}$

~~2. Displacement~~

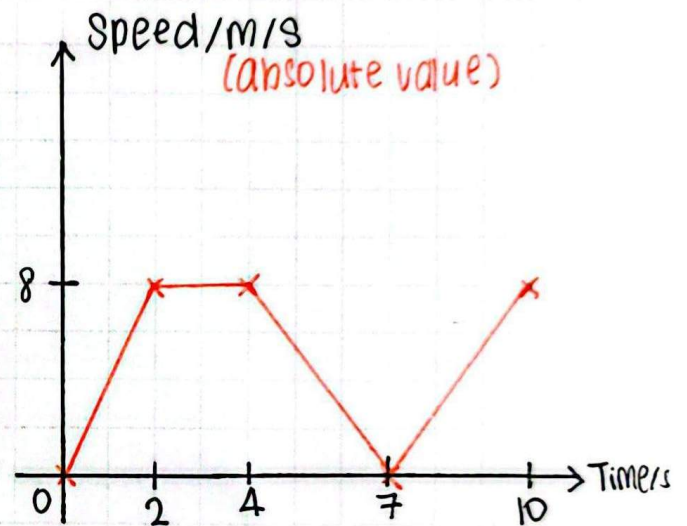
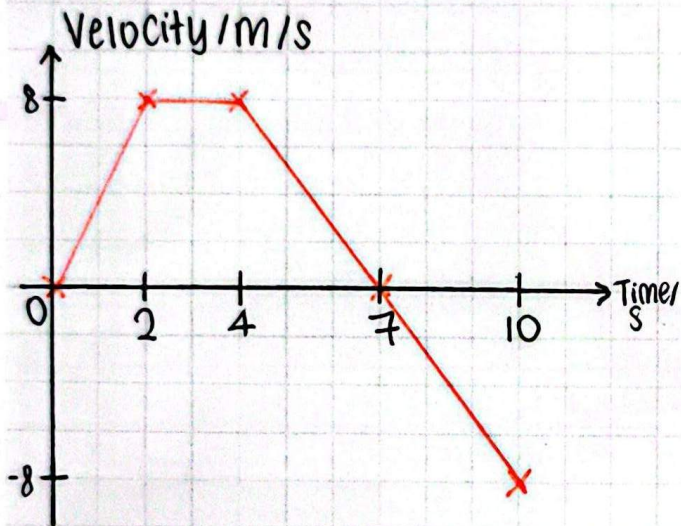
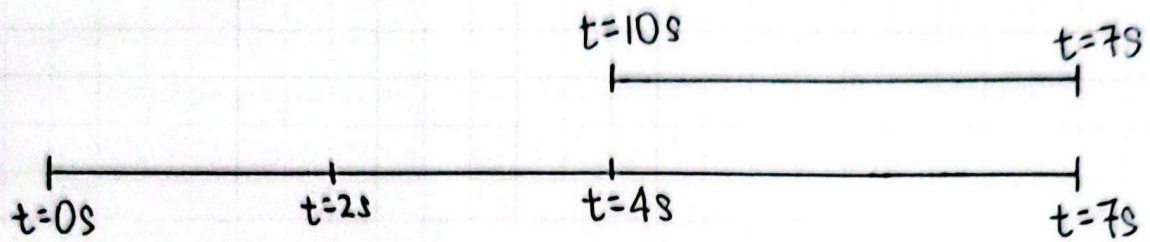
2. Average Velocity

Displacement $\rightarrow \sqrt{8^2+8^2} = 11.3\text{m (3.s.f.)}$

Average velocity $\rightarrow 11.3 \div 10 = 1.13\text{m/s}$

Direction $\rightarrow$ due North-East

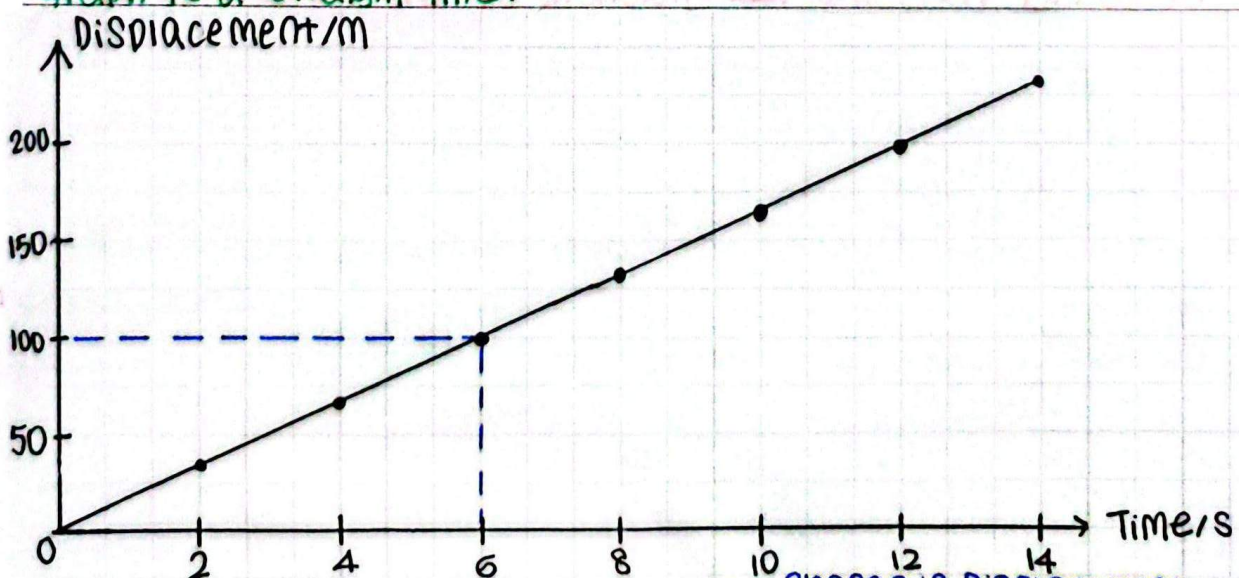
Speed-Time Graph vs Velocity-Time Graph



Interpreting Displacement-Time Graph

Object with constant velocity

- For an object moving with constant velocity, the displacement-time graph is a straight line.



$$\begin{aligned} \text{Change in displacement} \\ = 100 - 0 = 100\text{m} \end{aligned}$$

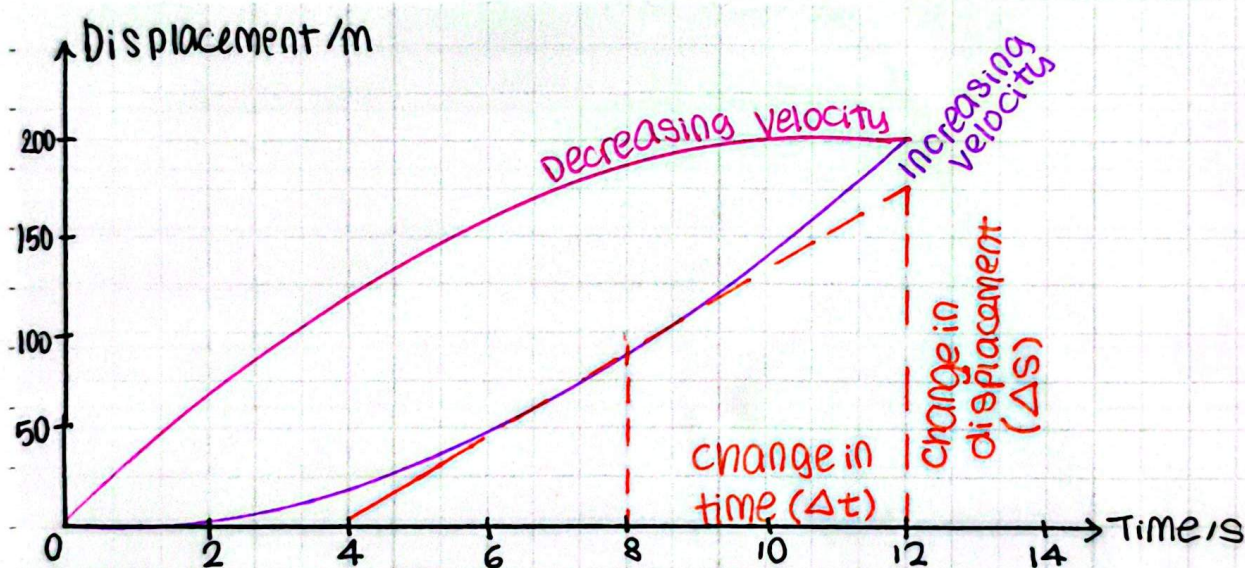
$$\begin{aligned} \text{Velocity} &= \frac{\text{change in Displacement}}{\text{Time Taken}} \\ &= \frac{100}{6} = 16.7\text{m/s} \end{aligned}$$

Object with Increasing velocity

- For an object moving with increasing velocity (acceleration), the displacement-time graph is a curve.
- Gradient to the tangent of the curve increases as time increases.

Object with Decreasing Velocity

- For an object moving with decreasing velocity (deceleration), the displacement-time graph is a curve.
- Gradient to the tangent of the curve decreases as time increases.



Finding Instantaneous velocity

- can be found from the ~~tan~~ gradient of the tangent at a point on the displacement-time graph.
- Example: (see graph above)

At $t = 8\text{s}$, the instantaneous velocity is

$$v = \frac{\text{change in displacement } (\Delta s)}{\text{time taken } (\Delta t)} = \frac{175}{8} = 21.9 \text{ m/s}$$

- This is equivalent to finding the gradient using gradient formula:

$$\text{Gradient} = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

Acceleration (definition)

- Acceleration is the rate of change of velocity
- When the car increases its speed, we say the car is accelerating.
- When the car decreases its speed, we say the car is decelerating.

Formula for acceleration

$$\text{Acceleration (m/s}^2\text{)} = \frac{\text{Final velocity} - \text{Initial velocity (m/s)}}{\text{Time taken (s)}}$$

$$\text{or } a = \frac{v - u}{t}$$

- When acceleration $> 0 \rightarrow$ acceleration
- When acceleration $< 0 \rightarrow$ deceleration
- Acceleration is a vector quantity.
- SI unit $\rightarrow \text{m/s}^2$

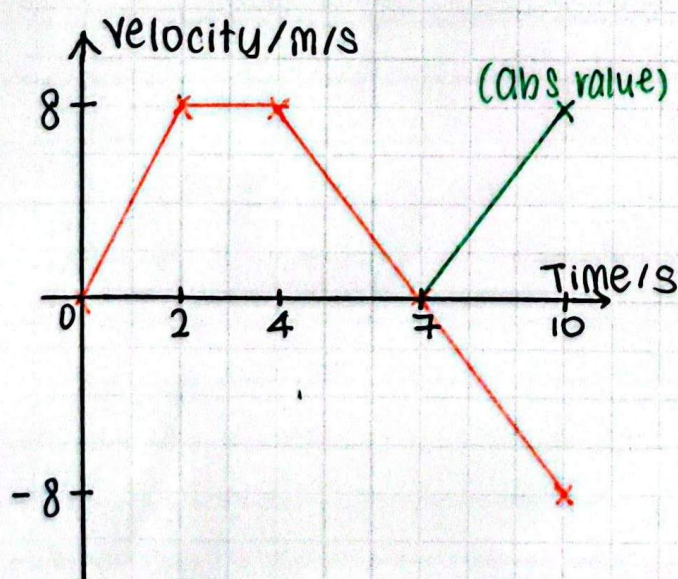
Interpreting acceleration

1. Positive acceleration, $\propto \text{m/s}^2$

$\rightarrow$ For every one second, the speed increases by $\propto \text{m/s}^2$.

2. Negative acceleration, $-\propto \text{m/s}^2$

$\rightarrow$ For every one second, the speed decreases by $\propto \text{m/s}^2$.



Accelerating or decelerating?

0 to 2s Accelerating (constant)

2 to 4s No acceleration/
constant velocity

4 to 7s Decelerating (constant)

7 to 10s Accelerating (constant)

★ Tip: Absolute value the graph (speed-time graph instead) for easier visualisation.

Zero acceleration

- When velocity is constant, there is zero acceleration.
(horizontal graph)

Constant acceleration / deceleration

- When velocity is increasing/decreasing at a constant rate, the object has a constant acceleration / deceleration.

Describe graph:

- It is a straight line graph with negative / positive gradient, which means object is decelerating / accelerating.

Increasing acceleration / deceleration

- When velocity is increasing/decreasing at an increasing rate, the object has an increasing acceleration / deceleration.

Describe graph:

- It is a graph with increasing ~~decreasing~~ gradient, which means the object is increasingly accelerating / decelerating.

Decreasing acceleration / deceleration

- When velocity is increasing/decreasing at a decreasing rate, the object has a decreasing acceleration / deceleration.

Describe graph:

- It is a graph with decreasing gradient, which means the object is decreasingly accelerating / decelerating.

Interpretation of Velocity-time Graph

Format:

The object moves with <acceleration> [from <time> to <time>] and velocity increases/decreases from <velocity> to <velocity>.

Description of acceleration:

- constant velocity
- constant/increasing/decreasing acceleration/deceleration

[] write only if time stamps are given.

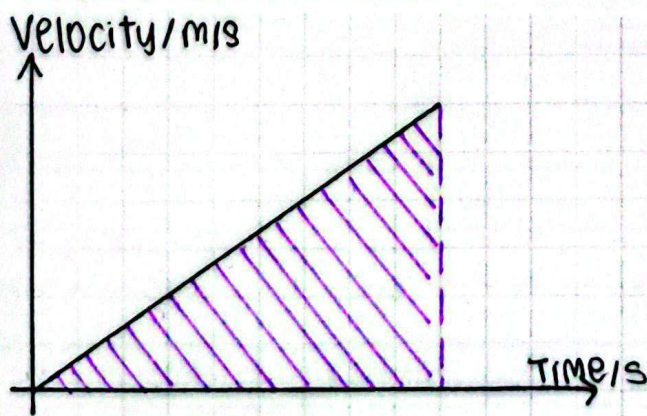
{ } write only for acceleration/deceleration (change in velocity)

Examples:

- The object is at rest / comes to a stop and is not moving.
- The object moves with increasing acceleration <acceleration> from time = 0.0s to time = 4.0s <time> and velocity increases from 0.0 m/s to 10.0 m/s <velocity>.

Displacement from velocity-time Graph

- The area under the velocity-time graph gives the displacement of object.



★ For negative velocities, absolute value the graph to find distance travelled.

For displacement, positive part(s) - negative part(s)

Acceleration due to gravity / acceleration of free fall (def)

- Acceleration of free fall is the rate of change in velocity due to the pull of gravity and it is ~~a~~ constant.

(Earth $\rightarrow 10 \text{ m/s}^2$, Moon $\rightarrow 1.6 \text{ m/s}^2$)

- Object decelerates when moving upwards, object accelerates when moving downwards.

- Motion is considered free fall when air resistance is negligible.

Upward motion

- When object is moving upwards, it will eventually come to a rest ($v = 0 \text{ m/s}$) and falls down.

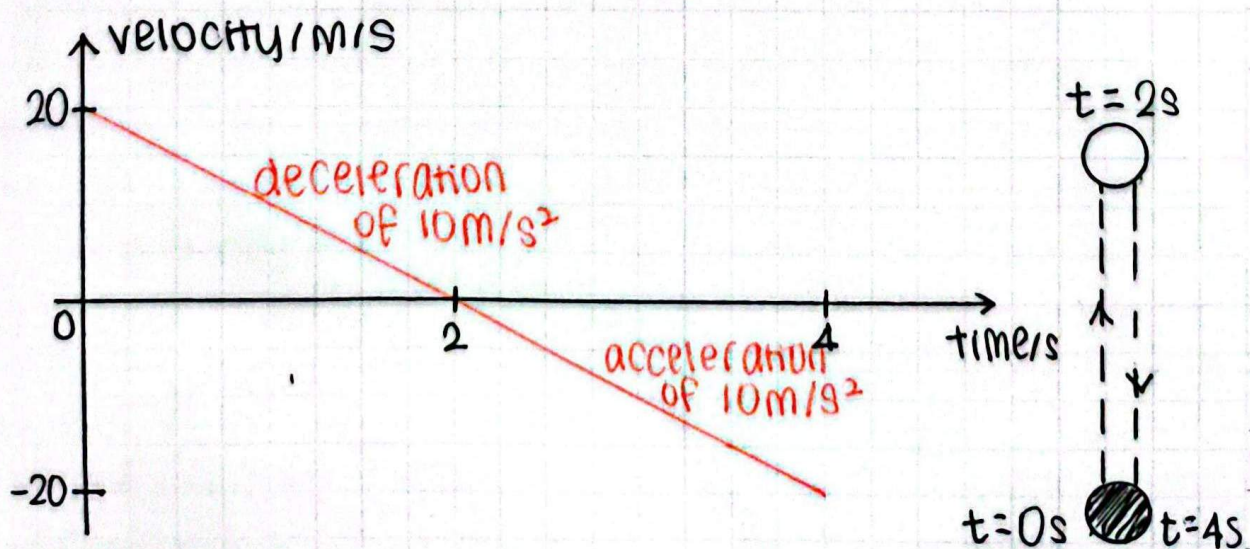
- At the highest point, acceleration = 10 m/s^2 (free fall)

- Constant deceleration

Downward motion

- Constant acceleration of 10 m/s^2 (free fall)

Velocity-time graph of upward & downward motion

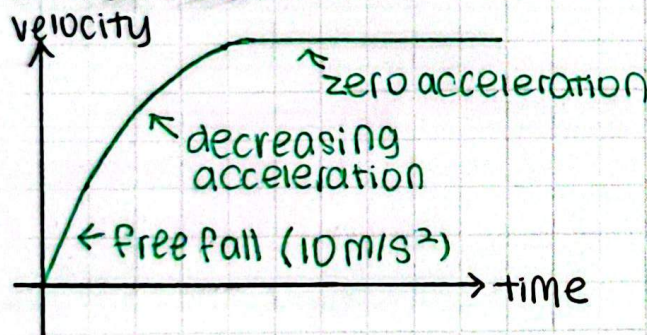


Air Resistance

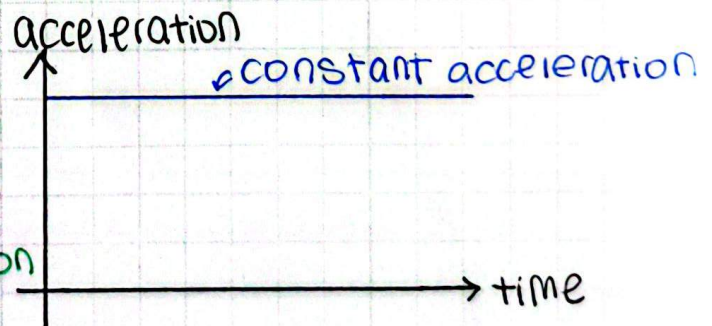
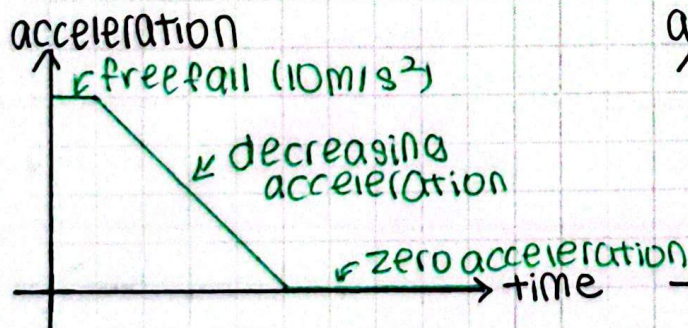
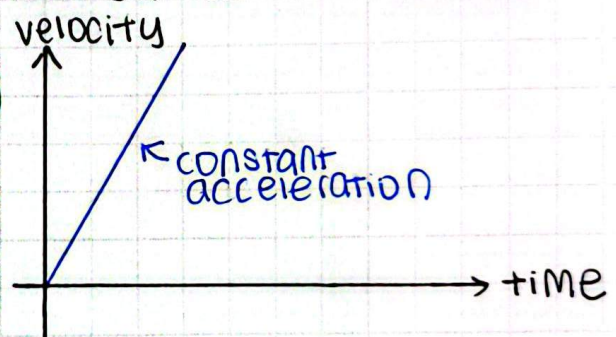
- In the presence of air resistance, a falling object will:
 - experience acceleration close to 10 m/s^2 when velocity is low
 - acceleration decreases as velocity increases
 - acceleration becomes zero and object continues to fall at constant velocity.
 - This constant velocity is known as terminal velocity.
- (chapter 4)

Object falling with air resistance vs no air resistance

with air resistance



without air resistance



★ Memorise descriptions of graph (description of motion) ★

Terminal velocity graph (air resistance)

